COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE212 - Automating Server Management	**SECTION:**	CPE31S2
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ACTIVITY NO. 2 **SSH Key-Based Authentication and GIT Setup**

1. PROCEDURE OUTPUT

Task 1: Create an SSH Key Pair for User Authentication

1. The simplest way to generate a key pair is to run ssh-keygen without arguments. In this case, it will prompt for the file in which to store keys. First, the tool asked where to save the file. SSH keys for user Authentication is usually stored in the users .ssh directory under the home directory. However, in enterprise environments, the location is often different. The default key file name depends on the algorithm, in this case id_rsa when using the default RSA algorithm. It could also be, for example, id_dsa or id_ecdsa.

2. Issue the command ssh-keygen -t rsa -b 4096. The algorithm is selected using the -t option and key size using the -b option.

3. When asked for a passphrase, just press enter. The passphrase is used for encrypting the key, so that it cannot be used even if someone obtains the private key file. The passphrase should be cryptographically Strong.

4. Verify that you have created the key by issuing the command ls -la .ssh. The command should show the .ssh directory containing a pair of keys. For example, id_rsa.pub and id_rsa.

```
quioyo3@WorkStation:~/Desktop$ ssh-keygen -t rsa -b 4096
Generating public/private rsa key pair.
Enter file in which to save the key (/home/quioyo3/.ssh/id_rsa):
Enter passphrase (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in /home/quioyo3/.ssh/id_rsa
Your public key has been saved in /home/quioyo3/.ssh/id_rsa.pub
The key fingerprint is:
SHA256:9eucmEzNN/RLUiOqDHEpjtGqFkVtVf18CFkibz2mRzI quioyo3@WorkStation
The key's randomart image is:
+---[RSA 4096]----+
|      . ..o.oo.  |
|     . o   oo+   |
|    . .    . E.B. |
|     ..  ..o *.+.|
|    .. +So  o.oo.|
|   .  = +  o.+o..|
|    .o o  ..+.o..|
|   ..   oo.= oo..|
|  ..     o+ +  . |
+----[SHA256]-----+
quioyo3@WorkStation:~/Desktop$ ls -la ~/.ssh
total 24
drwx------  2 quioyo3 quioyo3 4096 Jul 30 09:54 .
drwxr-x--- 15 quioyo3 quioyo3 4096 Jul 30 08:53 ..
-rw-------  1 quioyo3 quioyo3 3381 Jul 30 09:54 id_rsa
-rw-r--r--  1 quioyo3 quioyo3  745 Jul 30 09:54 id_rsa.pub
-rw-------  1 quioyo3 quioyo3 1956 Jul 30 09:02 known_hosts
-rw-------  1 quioyo3 quioyo3 1120 Jul 30 09:02 known_hosts.old
quioyo3@WorkStation:~/Desktop$
```

Task 2: Copying the Public Key to the remote servers

1. To use public key authentication, the public key must be copied to a server and installed in an authorized_keys file. This can be conveniently done using the ssh-copy-id tool.

2. Issue the command similar to this: ssh-copy-id -i ~/.ssh/id_rsa user@host

3. Once the public key has been configured on the server, the server will allow any connecting user that has the private key to log in. During the login process, the client proves possession of the private key by digitally signing the key exchange.

```
quioyo3@WorkStation:~/Desktop$ ssh-copy-id -i ~/.ssh/id_rsa quioyo1@192.168.56.109
/usr/bin/ssh-copy-id: INFO: Source of key(s) to be installed: "/home/quioyo3/.ssh/id_rsa.pub"
/usr/bin/ssh-copy-id: INFO: attempting to log in with the new key(s), to filter out any that are
already installed
/usr/bin/ssh-copy-id: INFO: 1 key(s) remain to be installed -- if you are prompted now it is to i
nstall the new keys
quioyo1@192.168.56.109's password:

Number of key(s) added: 1

Now try logging into the machine, with:   "ssh 'quioyo1@192.168.56.109'"
and check to make sure that only the key(s) you wanted were added.
```

```
quioyo3@WorkStation:~/Desktop$ ssh-copy-id -i ~/.ssh/id_rsa quioyo2@192.168.56.110
/usr/bin/ssh-copy-id: INFO: Source of key(s) to be installed: "/home/quioyo3/.ssh/id_rsa.pub"
/usr/bin/ssh-copy-id: INFO: attempting to log in with the new key(s), to filter out any that are
already installed
/usr/bin/ssh-copy-id: INFO: 1 key(s) remain to be installed -- if you are prompted now it is to i
nstall the new keys
quioyo2@192.168.56.110's password:

Number of key(s) added: 1

Now try logging into the machine, with:   "ssh 'quioyo2@192.168.56.110'"
and check to make sure that only the key(s) you wanted were added.
```

```
quioyo3@WorkStation:~/Desktop$ ssh quioyo1@192.168.56.109
Welcome to Ubuntu 22.04.5 LTS (GNU/Linux 6.8.0-124-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

Expanded Security Maintenance for Applications is not enabled.

116 updates can be applied immediately.
109 of these updates are standard security updates.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

New release '24.04.4 LTS' available.
Run 'do-release-upgrade' to upgrade to it.

Last login: Thu Jul 30 08:55:55 2026 from 192.168.56.108
quioyo1@server1:~$ exit
logout
Connection to 192.168.56.109 closed.
```

```
quioyo3@WorkStation:~/Desktop$ ssh quioyo2@192.168.56.110
Welcome to Ubuntu 22.04.5 LTS (GNU/Linux 6.8.0-136-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

New release '24.04.4 LTS' available.
Run 'do-release-upgrade' to upgrade to it.

Last login: Thu Jul 30 09:02:41 2026 from 192.168.56.108
quioyo2@server2:~$ exit
logout
Connection to 192.168.56.110 closed.
```

4. On the local machine, verify that you can SSH with Server 1 and Server 2. What did you notice? Did the connection ask for a password? If not, why?
-What I notice is that it logs directly without asking for password, maybe because the server verifies the identity using the public key stored in the authorized keys file.

Task 3: Set up the Git Repository
1. On the local machine, verify the version of your git using the command which git. If a directory of git is displayed, then you don't need to install git. Otherwise, to install git, use the following command: sudo apt install git

```
quioyo3@WorkStation:~/Desktop$ which git
/usr/bin/git
quioyo3@WorkStation:~/Desktop$ git --version
git version 2.34.1
quioyo3@WorkStation:~/Desktop$
```

```
quioyo3@WorkStation:~/Desktop$ which git
quioyo3@WorkStation:~/Desktop$ sudo apt update && sudo apt install git -y
[sudo] password for quioyo3:
Hit:1 http://ph.archive.ubuntu.com/ubuntu jammy InRelease
Hit:2 http://ph.archive.ubuntu.com/ubuntu jammy-updates InRelease
Hit:3 http://ph.archive.ubuntu.com/ubuntu jammy-backports InRelease
Hit:4 http://security.ubuntu.com/ubuntu jammy-security InRelease
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
109 packages can be upgraded. Run 'apt list --upgradable' to see them.
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following packages were automatically installed and are no longer required:
  libfwupd2 libfwupdplugin5 libgcab-1.0-0 libsmbios-c2
Use 'sudo apt autoremove' to remove them.
The following additional packages will be installed:
  git-man liberror-perl
Suggested packages:
  git-daemon-run | git-daemon-sysvinit git-doc git-email git-gui gitk gitweb git-cvs
  git-mediawiki git-svn
The following NEW packages will be installed:
  git git-man liberror-perl
0 upgraded, 3 newly installed, 0 to remove and 109 not upgraded.
Need to get 4,154 kB of archives.
After this operation, 21.0 MB of additional disk space will be used.
Get:1 http://ph.archive.ubuntu.com/ubuntu jammy/main amd64 liberror-perl all 0.17029-1 [26.5 kB]
Get:2 http://ph.archive.ubuntu.com/ubuntu jammy-updates/main amd64 git-man all 1:2.34.1-1ubuntu1.
17 [954 kB]
Get:3 http://ph.archive.ubuntu.com/ubuntu jammy-updates/main amd64 git amd64 1:2.34.1-1ubuntu1.17
 [3,174 kB]
Fetched 4,154 kB in 3s (1,454 kB/s)
Selecting previously unselected package liberror-perl.
(Reading database ... 203827 files and directories currently installed.)
Preparing to unpack .../liberror-perl_0.17029-1_all.deb ...
```

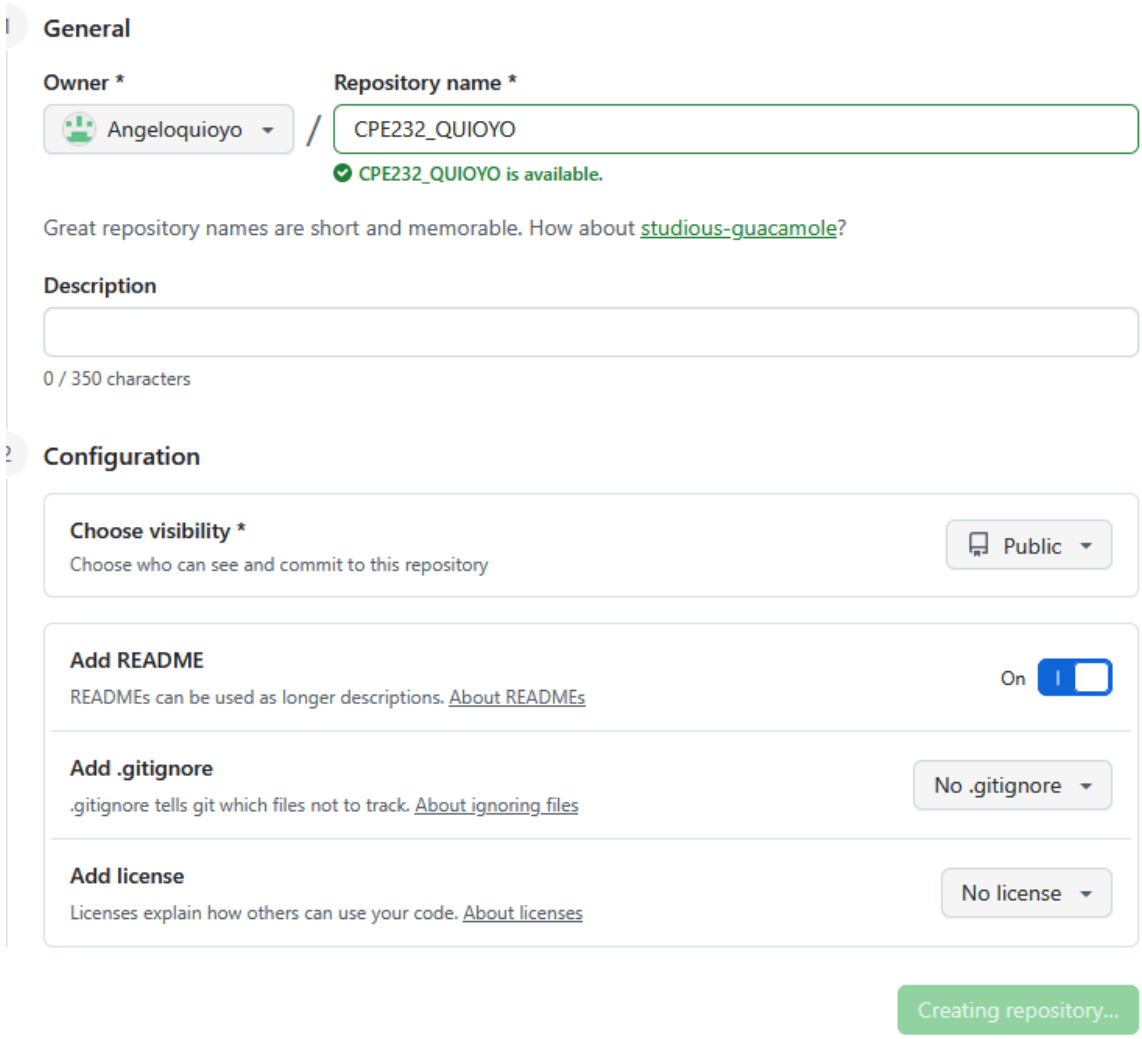
2. After the installation, issue the command which git again. The directory of git is usually installed in this location: user/bin/git.

3. The version of git installed in your device is the latest. Try issuing the command git --version to know the version installed.

4. Using the browser in the local machine, go to www.github.com.

5. Sign up in case you don't have an account yet. Otherwise, login to your GitHub account.

a. Create a new repository and name it as CPE232_yourname. Check Add a README file and click Create a repository.



b. Create a new SSH key on GitHub. Go to your profile's settings and click SSH and GPG keys. If there is an existing key, make sure to delete it. To create new SSH keys, click New SSH Key. Write CPE232 key as the title of the key.

c. On the local machine's terminal, issue the command cat .ssh/id_rsa.pub and copy the public key. Paste it on the GitHub key and press Add SSH key.

d. Clone the repository that you created. In doing this, you need to get the link from GitHub. Browse to your repository as shown below. Click on the Code drop down menu. Select SSH and copy the Link.

e. Issue the command git clone followed by the copied link. For example, git clone git@github.com:jvtaylar-cpe/CPE232_yourname.git. When prompted to continue connecting, type

yes and press enter.

f. To verify that you have cloned the GitHub repository, issue the command ls. Observe that you have the CPE232_yourname in the list of your directories. Use CD command to go to that directory and LS command to see the file README.md.

g. Use the following commands to personalize your git.
- git config --global user.name "Your Name"
- git config --global user.email yourname@email.com
- Verify that you have personalized the config file using the command cat ~/.gitconfig

h. Edit the README.md file using nano command. Provide any information on the markdown file pertaining to the repository you created. Make sure to write out or save the file and exit.

i. Use the git status command to display the state of the working directory and the staging area. This command shows which changes have been staged, which haven't, and which files aren't being tracked by Git. Status output does not show any information regarding the committed project history. What is the result of issuing this command?

j. Use the command git add README.md to add the file into the staging area.

k. Use the git commit -m "your message" to create a snapshot of the staged changes along the timeline of the Git projects history. The use of this command is required to select the changes that will be staged for the next commit.

l. Use the command git push <remote><branch> to upload the local repository content to GitHub repository. Pushing means to transfer commits from the local repository to the remote repository. As an example, you may issue git push origin main.

m. On the GitHub repository, verify that the changes have been made to README.md by refreshing the page. Describe the README.md file. You can notice how long the last commit was. It should be some minutes ago and the message you typed on the git commit command should be there. Also, the README.md file should have been edited according to the text you wrote.

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

1. How will you describe the ssh-program? What does it do?
   - Describing ssh-program it is a network protocol that is used to log into remote computers safely. It encrypts the connection over an insecure network. It allows you to run the terminal commands, transfer files, and manage remote servers
2. How do you know that you already installed the public key to the remote servers?
   - Testing the installation by logging in with SSH key authentication . Running the command ssh username@remote_host in the terminal. When installed correctly if the server logs you in without asking for the password. It can inspect remote servers.
3. What sort of things have we so far done to the remote servers using ansible commands?
   - Using Ansible ad-hoc commands and playbooks to automate administrative tasks. Testing its connectivity is by using the ping module. Install and update software packages. Manages system services like starting or restarting the web servers.

4. How important is the inventory file?
   - The inventory file defines the target servers ansible manages. It lists down IP addresses, domain names, and server groups. Ansible can't locate the servers without the file. It uses to assign custom variables, connection ports, and SSH users to a specific hosts.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: **Gemini**

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: **Use for guidance in putting the right command and to know what should the output be**

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: **It was only used for guidance in putting right codes and did not change or modify any code.**

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

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